

AGASTYA SETH

Staff AI Engineer | Agentic Systems, LLM Evals, AI Infrastructure

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EXPERIENCE

SYNOPSIS

Sunnyvale, CA

Staff AI Engineer, HAV R&D (Tech Lead, Agentic AI)

Jun 2025 - Present

- Lead HAV's agentic AI efforts for the customer-facing **AgentEngineer** and **Autopilot** products, driving cross-team architecture and delivery of task-level agents for compile convergence, system RCA, analytics, and performance; alpha releases rolling out to customers.
- Architected a long-running autonomous orchestrator that plans, dispatches, and monitors task agents across **20-30 hour** EDA tool runs; custom harnesses and plugins manage async processes, Slurm/batch jobs on the compute farm, and failure recovery.
- Built a secure, scalable agentic chat engine on **OpenCode** adopted by **~800 monthly users** across HAV; it queries multiple internal data sources through custom tools, triages issues, and auto-generates code reviews.
- Designed built-in security guardrails (scoped data access, sandboxed tool execution, prompt-injection and data-leakage safeguards) enabling safe agent use over proprietary source code and design data.
- Own agent evaluation strategy: building scalable eval pipelines and curating real-world designs and benchmark task suites to measure agent accuracy, reliability, and regressions across models and releases.

CADENCE DESIGN SYSTEMS

San Jose, CA

Machine Learning Intern, Cerebrus R&D

May 2024 - Aug 2024

- Built ML models (XGBoost, MLP) that predict timing (TNS) and total power directly from Verilog RTL using code-AST features, delivering design-quality estimates without full synthesis.
- Automated feature extraction and evaluation with unit tests; improved generalization to unseen designs by fixing data-quality issues.

CADENCE DESIGN SYSTEMS

Noida, India

Software Engineer II, Quantus R&D

Jul 2020 - Jul 2023

- Built high-performance C++ graph and computational-geometry algorithms for parasitic extraction across **10+** R&D projects; owned Rcompare, a comparison engine for multi-million-net designs, and hardened core libraries for performance and memory safety.
- Led early AI initiatives: an ML-based runtime ETA estimator and an LLM-powered query tool for engineering data, presented at internal conferences.

LIQVID / SHIV NADAR UNIVERSITY

Remote

NLP Research Scientist (part-time)

Aug 2020 - Sep 2021

- Built a transformer-based automated essay scoring system for publishers including Wiley, combining fine-tuned DistilBERT with an ensemble LSTM for explainability; achieved **0.82 kappa**, deployed on AWS SageMaker.

VISENZE

Singapore

Data Science Intern

Jan 2020 - Jul 2020

- Improved fashion-attribute CNN models' F1 by **15-20%** via augmentation, attention, and tuning; built a t-SNE-based data sufficiency pipeline that cut iteration time by **10x**.

SELECTED RESEARCH AND PUBLICATIONS

MALLS: Multi-Agent LLMs for Synthetic Hardware Vulnerability Generation and Detection	ICCD 2025
Focus Session: Do Agentic LLMs Change the Paradigm of Hardware Test Generation?	DATE 2026
UnSeenTimeQA: Time-Sensitive Question-Answering Beyond LLMs' Memorization	ACL 2025
The Art of Defending: Evaluating LLM Defense Strategies on Safety and Over-Defensiveness	Findings of ACL 2024
REACT: RL-Based Adaptive ECG Anonymization and Privacy Threat Mitigation	COINS 2025
LLM-DATA: LLM-Aided Design and Test Automation for Rapid Circuit and Systems Development	IEEE TCAD (under review)

EDUCATION

Arizona State University, M.S. Computer Science (GPA 4.0); Graduate Research Assistant, ACME Lab	May 2025
Shiv Nadar University, B.Tech. Electronics and Communication Engineering, Minor in Mathematics	Jun 2020

TECHNICAL SKILLS

GenAI and Agents: LLM agents, multi-agent systems, agent orchestration, agentic workflows, tool/function calling, MCP, RAG, vector databases (FAISS), prompt and context engineering, LangChain, LangGraph, OpenCode, Claude and OpenAI APIs, human-in-the-loop

Evaluation and Safety: LLM evals, benchmark design, LLM-as-a-judge, red-teaming, guardrails, prompt-injection defense, AI safety and alignment, observability, LLMOps, MLOps

ML and Training: PyTorch, JAX, TensorFlow, Hugging Face Transformers, fine-tuning (LoRA/PEFT), RLHF, DPO, PPO, reinforcement learning, diffusion and flow models, XGBoost, scikit-learn

Languages and Infra: Python, C++, SQL, JavaScript, Bash, Tcl, SystemVerilog, Docker, Kubernetes, AWS (SageMaker, EC2, S3), Slurm, Git, CI/CD, Linux

Awards: Smart India Hackathon 2019 Winner (300k participants); full-tuition Graduate Research Assistantship, ASU